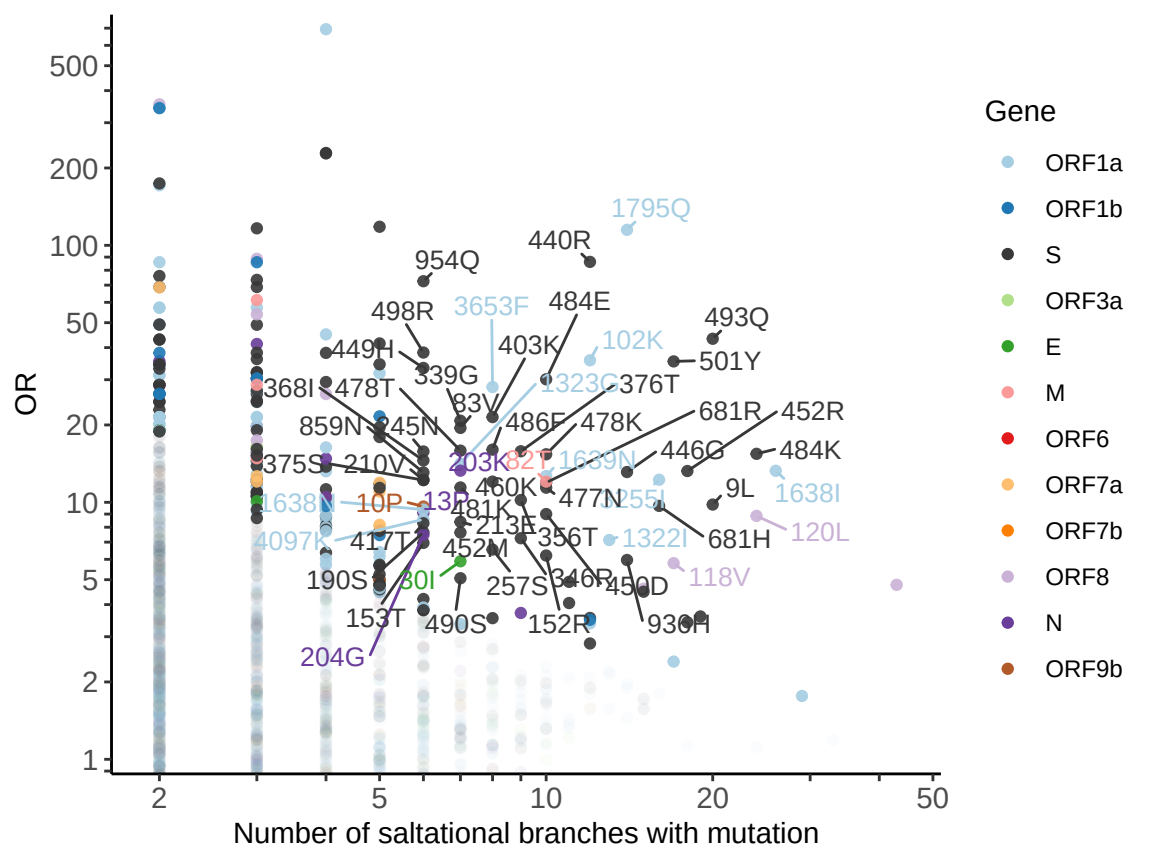

OR
500
200
100
50
20
10
5
2
1
2
5
10
20
50
Number of saltational branches with mutation
1795Q
440R
954Q
498R
3653F
484E
493Q
449H
403K
102K
501Y
339G
1323G
376T
368I
478T
83V
681R
452R
859N
245N
486F
478K
446G
484K
375S
210V
203K
82T
1639N
460K
9L
1638I
1638N
10P
13P
477N
481K
213E
120L
4097K
417T
452M
356T
1322I
681H
190S
80I
257S
346R
450D
118V
153T
490S
152R
930H
204G
Gene
ORF1a
ORF1b
S
ORF3a
E
M
ORF6
ORF7a
ORF7b
ORF8
N
ORF9b